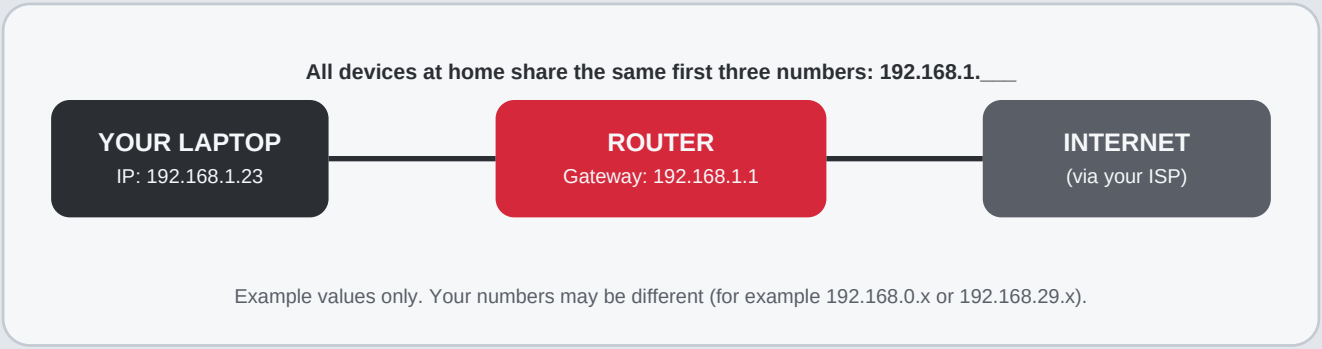
HANDS-ON LAB / 4 TASKS

IP Addressing Lab: Router Admin, IP Settings and DHCP

A step-by-step guide with fill-in worksheets. You do the steps on **your own router and device**, write your real values in the boxes and paste your screenshots into the marked spaces.



Example values only. Your numbers may be different (for example 192.168.0.x or 192.168.29.x).

Key words in easy language

Term	Meaning	Example
IP address	The address of your device on the network, like a house number.	192.168.1.23
Subnet mask	Tells which part of the IP is the network (street) and which is the device (house).	255.255.255.0
Default gateway	The door out of your network. It is almost always the router's own IP.	192.168.1.1
DHCP	The router's automatic system that gives each device an IP address.	Obtain IP automatically
Static IP	An IP address you type in yourself so it never changes.	192.168.1.50
DNS	Turns website names (google.com) into IP addresses.	8.8.8.8

Before you start

- Use **your own home network** and your own devices only. Do not do this on school, office or public Wi-Fi.
- Keep the router's default login handy. It is usually printed on a **sticker at the back or bottom** of the router.
- Write down your original settings first. In Task 3, always switch back to **automatic (DHCP)** when you finish.

1 Open the Router Login Page and Take a Screenshot

Find the router IP, open it in a browser and capture the login screen

Step 1: Find your router's IP (default gateway)

```
C:\> ipconfig

Wireless LAN adapter Wi-Fi:
IPv4 Address. . . . : 192.168.1.23
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . : 192.168.1.1 <-- this is your router
```

Device	How to find the Default Gateway
Windows	Press Windows + R, type **cmd**, press Enter, then type **ipconfig** and look for Default Gateway.
Mac	System Settings, then Wi-Fi, then Details, then TCP/IP. Look at the Router line.
Android	Settings, then Wi-Fi, tap your network (or its gear icon). Look at Gateway. On some phones tap Advanced.
iPhone	Settings, then Wi-Fi, tap the (i) next to your network. Look at Router.

Step 2: Open it in a browser

1	Open Chrome or Edge and type the gateway in the address bar, for example **http://192.168.1.1**, then press Enter.
2	The router **login page** appears (username and password boxes). Do not change anything.
3	Take a screenshot with the **address bar visible** so your teacher can see the IP. Windows: Win + Shift + S. Phone: power + volume down.

PASTE YOUR LOGIN PAGE SCREENSHOT HERE

Make sure the router IP in the address bar is visible

Common router IPs: 192.168.0.1, 192.168.1.1, 10.0.0.1 and 192.168.29.1 (often seen on Jio routers). If one does not open, use the one from ipconfig. If the page says "not secure", that is normal for home routers.

2 Note Your Current IP, Subnet Mask and Default Gateway

Check these while your laptop or phone is connected to your home Wi-Fi

How to find them

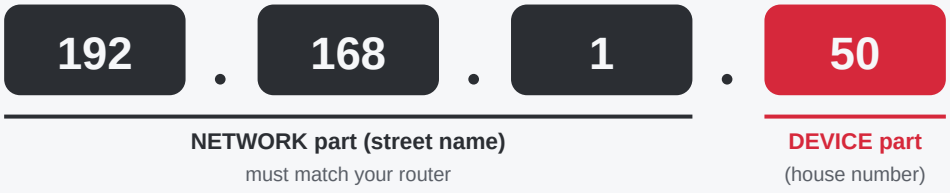
Device	Steps
Windows laptop	Open Command Prompt and run **ipconfig**. Under *Wireless LAN adapter Wi-Fi* read IPv4 Address, Subnet Mask and Default Gateway. For extra detail run **ipconfig /all**.
Android phone	Settings, Wi-Fi, tap your connected network. Open Advanced or View more to see IP address, Gateway and Network prefix length (24 means 255.255.255.0).
iPhone	Settings, Wi-Fi, tap the (i) next to your network. See IP Address, Subnet Mask and Router.

My values (fill in)

Item	Example	Your value
Device used	Example: Dell laptop	_____
IP address	192.168.1.23	_____
Subnet mask	255.255.255.0	_____
Default gateway	192.168.1.1	_____
DNS server (optional)	192.168.1.1	_____
Connected to (Wi-Fi name)	HomeWiFi	_____

Understand what you wrote

Example IP address 192.168.1.50 with subnet mask 255.255.255.0



- **Subnet mask 255.255.255.0** means the first three numbers (192.168.1) are the network and the last number is the device. That allows up to **254 devices** (1 to 254).
- **Gateway** should be the router's IP. If it is different from the address you opened in Task 1, check again.

3 Set a Static IP, Test Internet, Then Switch Back

Change from DHCP to a manual IP in the same range as your router

Rule: a static IP must share the **first three numbers** with the router. If the router is 192.168.1.1, use 192.168.1.**50** with mask 255.255.255.0. Pick a number that no other device uses (see Task 4 for the DHCP range).

Windows steps

1	Press Windows + R, type **ncpa.cpl** and press Enter. Right-click **Wi-Fi**, then Properties.
2	Select **Internet Protocol Version 4 (TCP/IPv4)**, then click Properties.
3	Choose **Use the following IP address** and enter the values from the table below. Choose **Use the following DNS server addresses** and enter DNS.
4	Click OK, then Close. Wait a few seconds for the connection to refresh.

Field	Example value (router = 192.168.1.1)
IP address	192.168.1.50
Subnet mask	255.255.255.0
Default gateway	192.168.1.1 (your router)
Preferred DNS	8.8.8.8 (or 192.168.1.1)

Test the internet

```
C:\> ping 192.168.1.1 (reaches the router?)
C:\> ping 8.8.8.8 (reaches the internet?)
C:\> ping google.com (DNS working?)
```

Also open a website in your browser. **Replies = working.** If *Request timed out* or pages do not load, re-check the gateway, mask and that the IP is not used by another device.

Phone steps

- **Android:** Wi-Fi, tap your network, Edit (pencil), Advanced options, IP settings: change **DHCP** to **Static**, then enter the values.
- **iPhone:** Wi-Fi, tap (i), Configure IP, change **Automatic** to **Manual**, then enter the values.

Go back when done (important): open the same IPv4 window, choose **Obtain an IP address automatically** and **Obtain DNS server address automatically**, then click OK. On phones choose DHCP / Automatic again. Run **ipconfig** to confirm the IP changed back.

3 My Static IP Test Results (fill in)

Record what you tried and what happened

Check	Example	My result
Static IP I used	Example: 192.168.1.50	_____
Ping to router worked?	Yes / No	_____
Ping to 8.8.8.8 worked?	Yes / No	_____
Website opened?	Yes / No	_____
Reverted to DHCP?	Yes, done	_____

Possible result: *"Internet worked with the static IP because the IP, mask, gateway and DNS were all correct and in the router's range. After testing I switched back to automatic."* Write your own result honestly.

4 Find the DHCP Range and Explain an IP Conflict

Router admin panel: LAN / DHCP settings. View only, do not change anything.

Where to look

1	Log in to the router (use the sticker or your ISP's default login). If you cannot, ask the person who manages the router.
2	Open **LAN**, **Network** or **Advanced**, then **DHCP Server** (the name changes by brand).
3	Note the **Start IP**, **End IP** and **Lease time**. Take a screenshot if allowed. Leave without clicking Save.

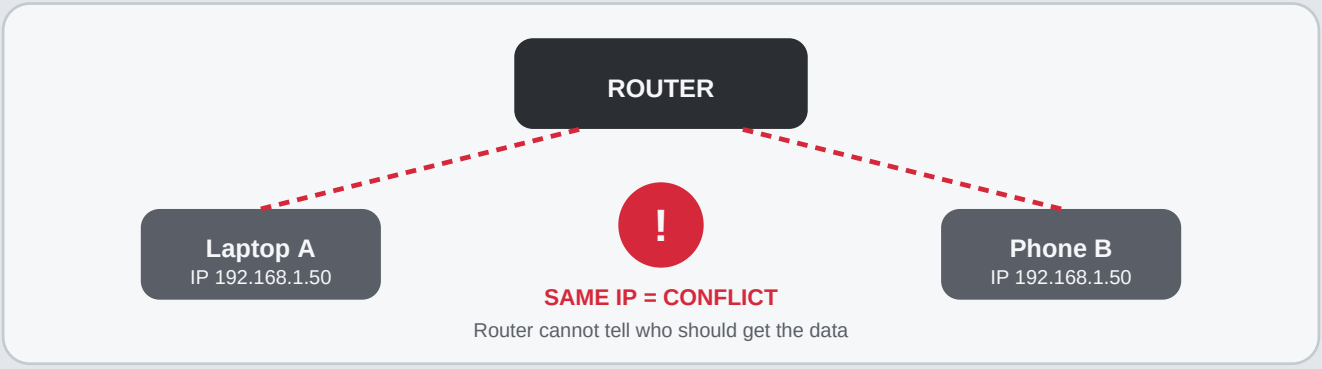
Setting	Example	My router
DHCP start IP	Example: 192.168.1.100	_____
DHCP end IP	Example: 192.168.1.199	_____
Lease time	Example: 24 hours	_____

PASTE DHCP SETTINGS SCREENSHOT HERE (optional)

Hide any passwords before sharing

4 What If Two Devices Get the Same IP?

Answer in 2-3 lines, plus a simple picture



My answer (2-3 lines)

If two devices are given the same IP address in the DHCP range, the router cannot tell which device should receive the data, so this is called an **IP conflict**. Both devices may lose internet, disconnect again and again, or show an "IP address conflict" warning. DHCP normally prevents this by giving every device a different address from the pool, so conflicts mostly happen when someone types a static IP that is already in use.

How to avoid conflicts

- Keep phones and laptops on **automatic (DHCP)** so the router hands out unique addresses.
- If you need a fixed IP, choose one **outside the DHCP range** (for example 192.168.1.50 when the range is .100 to .199), or use **DHCP reservation** in the router.
- If you see a conflict, switch the device to automatic, or disconnect and reconnect to Wi-Fi to get a new address.

Why 192.168.1.50 is a safe choice in the example

Address	Who uses it	Safe?
192.168.1.1	Router (gateway)	No, already taken
192.168.1.50	Not in DHCP pool (.100 to .199)	Yes, if no other device uses it
192.168.1.150	Inside DHCP pool	Risky, a phone may get it

Submission checklist

- Task 1: login page screenshot with the router IP visible.
- Task 2: IP, subnet mask and default gateway written down.
- Task 3: static IP test result written, and device set back to DHCP.
- Task 4: DHCP range written and conflict explained in 2-3 lines.